Digital Literacy Project: Final Comprehensive Report

Course: Digital Literacy (CSE1021)

University: VIT Bhopal

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Role: Student Digital Ambassador

Module 1: Digital Literacy Awareness Infographic (Canva)

1.1 Objective

The goal of Task 1 was to design a visual resource using **Canva** to define Digital Literacy and its essential components for the modern student.

1.2 Design Methodology

- **Template Selection:** Chose a professional infographic template to balance text and visuals.
- **Core Content:** Defined Digital Literacy, highlighted essential tools (GitHub, LinkedIn, Google Workspace), and outlined safe internet practices.
- **Branding:** Used a consistent color palette to maintain a professional "Digital Ambassador" aesthetic.

1.3 Reflection

"To complete the first module of this project, I utilized **Canva**, a versatile graphic design platform, to create a comprehensive Digital Literacy Awareness Infographic. My design is structured to serve as a visual roadmap for new engineering students at VIT Bhopal. It begins by defining digital literacy in a modern context, followed by a curated list of essential technical tools—specifically **GitHub** for version control, **LinkedIn** for professional networking, and **Google Workspace** for cloud collaboration. The final section of the infographic emphasizes 'Digital Hygiene,' outlining critical safe internet practices to protect one's personal data.

One aspect I found particularly interesting during the design process was the challenge of **data visualization**—specifically, how to distill complex cybersecurity concepts into simple, actionable icons and short bullet points without losing the core message. It was fascinating to see how a professional colour palette and consistent typography could transform a wall of technical text into an engaging resource. Balancing the 'Do’s and Don’ts' of online behavior while maintaining a positive, encouraging tone for my peers was a rewarding creative exercise that highlighted the importance of visual communication in the tech industry."

Module 2: Building the Student Digital Portfolio

2.1 Platform Selection & Technical Utility

To establish a multi-faceted professional identity, I optimized my presence across three industry-standard platforms. Each serves a specific strategic purpose in my growth as a Computer Science and Engineering student:

- **LinkedIn (Professional Networking & Branding):** I utilized LinkedIn to build a "Digital Resume" that transcends a static document. By crafting a professional headline and a detailed "About" section, I have positioned myself to connect with industry leaders and VIT Bhopal alumni. Over the next four years, I will use this platform to share project milestones, certifications, and internship experiences, building a visible "digital trail" of my professional evolution.
- **GitHub (Technical Laboratory & Version Control):** GitHub serves as my primary repository for all coding projects. Beyond just hosting files, I am using GitHub to practice **Git version control**, which is a mandatory skill in modern software development. By organizing my Digital Literacy tasks into structured folders and using meaningful commit messages, I am demonstrating my ability to manage a codebase—a crucial requirement for future technical interviews and collaborative open-source contributions.
- **Kaggle (Data Science & Analytical Research):** I chose Kaggle as my third platform to immerse myself in the data science ecosystem. As AI and Machine Learning become central to the CSE curriculum, Kaggle provides me with access to global datasets and competitive "Notebooks." I plan to use this platform to transition from a consumer of data to a contributor, participating in community challenges that sharpen my analytical thinking and predictive modeling skills.

2.2 Long-Term Digital Strategy (The Four-Year Roadmap)

My strategy for these platforms is centered on "Incremental Growth."

1. **Phase 1 (Freshman Year):** Focus on building the foundation by documenting academic assignments on GitHub and expanding my LinkedIn network within the university.
2. **Phase 2 (Sophomore Year):** Begin participating in Kaggle "Getting Started" competitions and contributing to small open-source projects on GitHub to understand collaborative workflows.
3. **Phase 3 (Junior Year):** Utilize the built-up credibility on LinkedIn to apply for internships like the Google STEP program, using my GitHub portfolio as live evidence of my coding proficiency.
4. **Phase 4 (Senior Year):** Showcase a complete, four-year technical history to recruiters, moving from basic classroom exercises to complex, independent engineering solutions.

2.3 Reflection: Building a Multi-Platform Digital Portfolio

"To establish a comprehensive professional identity, I chose to optimize my profiles on **LinkedIn**, **GitHub**, and **Kaggle**. Each platform serves a distinct purpose in my development as a Computer Science professional. **LinkedIn** acts as my digital resume and networking hub, allowing me to connect with industry leaders and track internship opportunities. **GitHub** is my technical laboratory, where I use version control to host source code and collaborate on open-source projects. Finally, I chose **Kaggle** as my third platform to explore data science, participate in machine learning competitions, and access high-quality datasets for my academic research.

Over the next four years, I plan to use these platforms as a dynamic record of my growth. On **GitHub**, I will document my journey from basic Python scripts to complex full-stack applications. On **Kaggle**, I aim to move from a 'Novice' to a 'Contributor' by participating in hackathons that sharpen my analytical skills. Simultaneously, I will use **LinkedIn** to share my certifications and project milestones, building a 'digital trail' that demonstrates my consistency to future recruiters at companies like Google or Microsoft. By maintaining this three-pillar portfolio, I am ensuring that my technical skills and professional network grow in parallel throughout my BTech journey."

Module 3: Technical Platforms & Collaboration

3.1 Part A: HackerRank (Problem Solving)

I completed the **'Solve Me First'** challenge to demonstrate algorithmic logic.

- **Process:** Implemented a function to calculate the sum of two integers, ensuring all test cases passed.
- **Outcome:** Successfully validated my ability to work within a competitive coding environment.

3.2 Part B: Google Forms & Sheets (Collaboration)

Designed a **Digital Literacy Awareness Quiz** to test peer knowledge.

- **Features:** Used 5 diverse questions (Multiple Choice & Short Answer) and linked the form to a **Google Sheet** for real-time data tracking.

3.3 Reflection: Technical Problem Solving and Collaborative Tools

"To complete the third module of this project, I utilized two distinct yet equally vital platforms: HackerRank for algorithmic problem solving and Google Workspace (specifically Google Forms and Sheets) for digital collaboration. On HackerRank, I practiced fundamental programming logic by successfully solving the 'Solve Me First' challenge. This allowed me to refine my ability to handle standard input/output and pass automated test cases—a core skill for any software developer. Simultaneously, I built a Digital Literacy Awareness Quiz using Google Forms, which I integrated with a Google Sheet to automate real-time data collection and response analysis.

These tools will be indispensable for my academic journey over the next four years at VIT Bhopal. HackerRank will serve as my primary training ground for technical interviews and competitive coding, helping me maintain a sharp logical edge for campus placements. On the other hand, mastering Google Workspace ensures that I can collaborate efficiently on group projects and manage large datasets for my research assignments. By bridging the gap between individual coding and collaborative data management, this module has equipped me with the dual-competency required to excel in both the academic and professional spheres of Computer Science."

Module 4: Professional Communication & Etiquette

4.1 Task Overview

This module focused on the "Digital Handshake"—the art of professional correspondence. I drafted two high-stakes emails:

1. **Academic:** Requesting a UHV assignment extension due to health issues.
2. **Industry:** An internship inquiry to a Google Recruiter for the STEP program.

4.2 Social Media Etiquette

I developed a **10-point 'Do's and Don'ts'** guide to help students manage their digital footprint and avoid unprofessional behavior that could alert recruiters.

4.3 Reflection: The Impact of Poor Digital Communication

"A significant real-world problem often arises in professional environments when 'High-Stakes' information is communicated via informal channels with poor tone. A common scenario is when a project lead sends a brief, all-caps message like 'NEED REPORT NOW' to a subordinate via an instant messaging app without context or a professional greeting. This lack of digital etiquette often leads to a 'Negativity Bias,' where the receiver interprets the message as a sign of extreme anger or a workplace crisis, causing unnecessary stress and a breakdown in team morale.

To avoid this, the communication should have been handled differently by using a formal email with a clear subject line and a structured body. A professional greeting, followed by a specific deadline and a brief explanation of the urgency, would have ensured clarity. Using proper 'Email Etiquette'—as practiced in Task 4—replaces ambiguity with professionalism, ensuring that the message is received as a task-oriented request rather than a personal or professional attack."

Module 5: Cybercrime Case Study (RTO Scam)

5.1 Part A: Case Study Analysis

"The 'RTO E-Challan' scam is a sophisticated form of Smishing and Social Engineering. In the Gondia/Bhandara incident, the crime began with an urgent SMS claiming the victim had a pending traffic fine. The message contained a 'spoofed' URL that mimicked the official Government of India Parivahan portal. Once the victim clicked the link, they were prompted to enter their vehicle details followed by sensitive banking credentials to 'clear the fine.' By gaining access to this data, the attackers bypassed security layers and instantly debited ₹5 lakh. This case demonstrates that cybercriminals exploit the fear of legal authority to bypass technical firewalls. For students, the primary targets are those unfamiliar with official government domain structures (.gov.in), making awareness the first line of defense."

5.2 Part B: Stay Safe Online Checklist

Developed an **8-point safety guide** for Indian students, highlighting the "Golden Rule" of UPI safety (PIN is only for sending) and national reporting channels (**1930**).

5.3 Reflection: Cyber Awareness and Behavioral Change

"What surprised me most while researching the RTO E-Challan scam in Maharashtra was the sheer psychological precision of the attackers. I was struck by how cybercriminals successfully bypassed advanced technical security layers not through 'hacking' code, but by exploiting a victim's instinctual fear of government authority and legal penalties. Seeing a single malicious link result in a ₹5 lakh loss in the Gondia region made me realize that technical firewalls are only as strong as the user's digital literacy.

As a direct result of this research, I have personally committed to changing my habit of 'passive trust' in SMS notifications. Previously, I would often click on shortened links out of curiosity or a sense of urgency. Moving forward, I will strictly follow a 'Manual Verification' protocol—always navigating directly to official .gov.in portals rather than interacting with any unsolicited message. This shift from reactive clicking to proactive verification is a fundamental change in

my digital hygiene that I will also advocate for as a Student Digital Ambassador."

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6. Final Project Conclusion and Learning Outcomes

6.1 Project Synthesis

The completion of these five comprehensive modules marks a significant milestone in my journey as a First-Year Computer Science and Engineering student at VIT Bhopal. This project was not merely a series of academic tasks, but a strategic roadmap designed to build a robust, professional, and secure digital identity. From the creative challenges of designing a Canva infographic to the analytical depth required for the RTO cybercrime case study, each task has contributed to a well-rounded understanding of what it means to be truly "Digitally Literate" in a globalized technical landscape.

6.2 Key Learning Outcomes

Through this project, I have achieved the following core competencies:

- **Technical Versatility:** I have mastered the foundational tools of a modern developer, including **GitHub** for version control, **HTML/CSS** for web presence, and **Kaggle** for future data-driven research.
- **Professional Communication:** I have refined my ability to navigate high-stakes academic and industry correspondence, ensuring that my "Digital Handshake" is as professional as my technical skill set.
- **Collaborative Problem Solving:** By utilizing **HackerRank** and **Google Workspace**, I have learned to solve algorithmic problems individually while managing data and collaborative projects in a cloud-based environment.
- **Cyber Resilience and Advocacy:** The deep-dive into the RTO E-Challan scam taught me that technical security is incomplete without human vigilance. I am now equipped to identify phishing vectors and utilize national reporting channels like **1930** and **cybercrime.gov.in**.

6.3 The Path Forward as a Digital Ambassador

As a **Student Digital Ambassador**, my responsibility extends beyond my own screen. The knowledge I have gained regarding digital hygiene, UPI safety, and

professional branding will be shared with my peers and the wider university community. I plan to continue updating my GitHub repository with my coursework and maintain an active LinkedIn presence to bridge the gap between my academic learning and industry expectations.

This project has effectively transformed my perspective on technology—from a tool for consumption to a platform for professional creation and community safety. I am now prepared to navigate the next four years of my B.Tech journey with the digital confidence and ethical responsibility required of a modern engineer